

**Design and Implementation of an MSP430-Based 3S Lithium-Ion Portable Power Supply
with Selectable-Output Buck Converter**

Engineering Design Report

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Executive Summary

This report documents the design and development of a portable, rechargeable power supply based on a 3S lithium-ion battery and a MSP430G2553 microcontroller. The primary objective was to create a device capable of producing selectable 3.3V 5V and 9V outputs in the absence of a bench top power supply. A second objective was to gain practical experience designing a switching power converter from individual components rather than relying on a preassembled DC-DC converter module.

The development of this project involved many engineering design tools. LTspice was used to simulate the buck converter and evaluate its expected behavior, Altium Designer was used for schematic capture and four-layer PCB design, Onshape was used to create the enclosure and Texas Instruments CCSTUDIO was used to write the firmware for the MSP430.

The power-conversion stage uses a TL494 PWM controller, IRS2110 gate driver, external power MOSFET, 68 μ H inductor, catch diode, filtering capacitors, and selectable feedback network. The system also incorporates an internal 3S balance charging module, relay-controlled charging connections, voltage, current, and temperature monitoring, mode and voltage selection switches, an OLED display and an illuminated enable button inside of a 3D printed enclosure. The MSP430 manages the system's OFF, OUTPUT and CHARGE operating modes while preventing the use of the charging and voltage output systems simultaneously. It also processes sensor measurements, updates the display, and disables the buck converter if overcurrent or overtemperature conditions are detected.

The completed prototype successfully produced all three intended output voltages. Multimeter testing measured central values of 3.30 V, 5.00 V, and 9.00 V, with observed variations of approximately ± 0.04 V, ± 0.03 V, and ± 0.07 V, respectively. These values agreed with the measurements displayed on the OLED. The outputs were demonstrated using both LEDs and a 380 brushed DC motor, confirming operation with low-current and practical inductive loads.

Hardware testing confirmed the successful operation of all major system functions and identified improvements for a future revision. These include the correction of the catch-diode footprint and replacing the ACS758 with a lower-range battery-current sensor to improve measurement accuracy at normal operating currents.

Although the full 2.5A continuous-output target, efficiency and ripple could not be fully tested with available equipment the project achieved its primary functional objectives. The resulting prototype successfully demonstrates the integration of power electronics, analog sensing, embedded programming, four-layer PCB design, battery management, hardware troubleshooting, user-interface development, and CAD design.

Introduction

The primary motivation for this project was to create a portable power supply for building and testing electronics projects. Many electronic circuits require standard supply voltages such as 3.3 V, 5 V, or 9 V, but a benchtop power supply is not always available or convenient. A rechargeable system with selectable voltage outputs would provide a practical tool that could be transported and used in a variety of work environments.

Rather than relying on a preassembled DC-DC converter module the power conversion stage was designed using individual components to aid in gaining a more robust understanding of power-electronics design. The development of this circuit demanded an application of switching-converter principles, component-selection calculations, simulation, and mixed-signal PCB-layout techniques.

More broadly, the project provided an opportunity to build upon previous experience with circuit design, embedded programming, and PCB development while gaining experience integrating electrical, software, and mechanical subsystems into a complete functioning device. The completed system was intended to combine power conversion, battery charging, system monitoring, user control, and status display function

The project's main objectives include:

1. Simulate design and build a custom buck converter rather than use a preassembled power-conversion module.
2. Integrate the battery and charging hardware so that the battery could remain installed during both charging and normal operation.
3. Design a MSP430 microcontroller-based control and monitoring system capable of managing the operating states, collecting electrical and thermal measurements
4. Implement the electrical design on a four-layer PCB
5. Develop the firmware required to present information to the user and work alongside hardware level safety precautions.
6. Design an enclosure that integrates the PCB, battery, charger, display, controls, indicators, and external connections into a complete portable device.

System Architecture

The power-management system is divided into five primary functional areas: battery charging, logic power, buck conversion, measurement, and embedded control. A rechargeable 3S lithium-polymer battery serves as the main energy source. The battery supplies the custom buck converter during output operation and remains installed inside the device while being charged through the integrated pass-through charging interface.

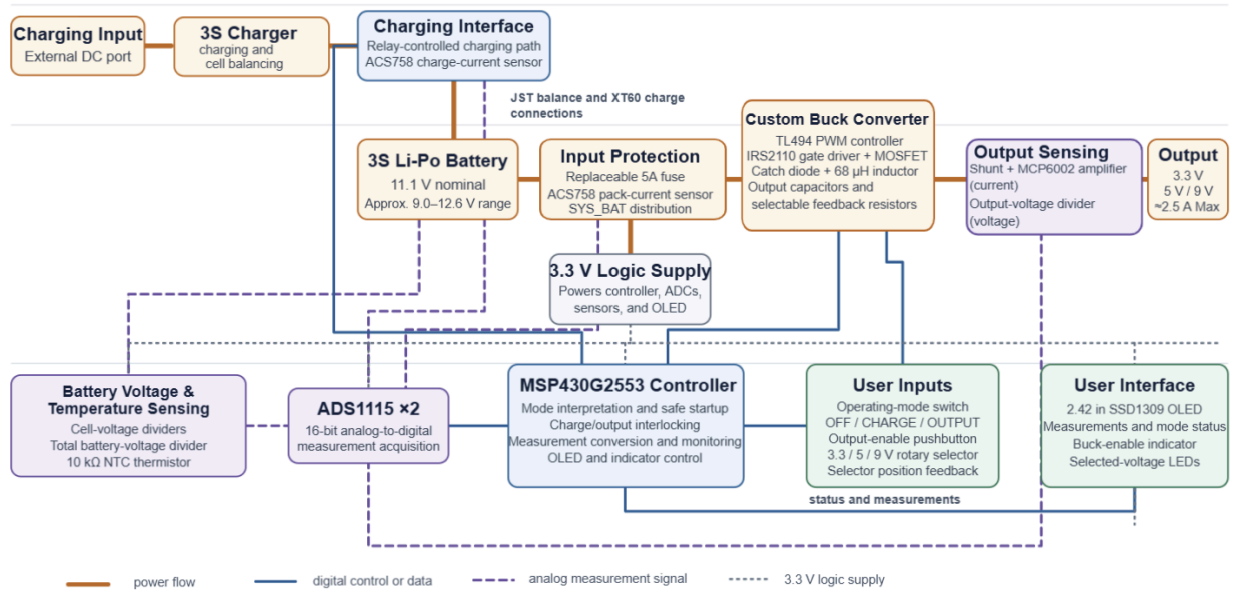


Figure 1: Overall system block diagram

Mode	Charge path	Buck converter	Logic and display	function
OFF	Disconnected	Disabled	Off	System shutdown
CHARGE	Connected through relays	Disabled	Active	Charge and monitor the battery
OUTPUT	Disconnected	Available	Active	Supply and monitor the selected output voltage

Figure 2: Operating Modes

The system has 3 main states selected by a physical operating mode switch. In CHARGE mode, relay switches connect the dedicated 3S valence charger to the battery while the buck converter remains disabled. In OUTPUT mode, the charging path is disconnected and buck can be enabled. The buck converter is initially disabled and is activated by an output-enable control signal. In OFF mode, both the charging and output functions are disabled.

A regulated 3.3 V logic rail supplies the MSP430G2553 microcontroller, ADS1115 analog-to-digital converters, measurement circuits, and OLED display during active operation. The MSP430 interprets the mode-selector and output-enable inputs, controls the charging and buck-enable signals, processes the acquired measurements, and updates the user interface. The measurement system monitors battery-cell voltage, total battery voltage, output voltage, battery and buck output current, charging current, and temperature. Relevant values and operating-state information are then presented on the OLED.

The system architecture intentionally separates the high-current power-conversion paths from the low-voltage sensing and control circuitry. This separation allows the MSP430 to supervise the system without carrying load current directly. A physical mode switch prevents the charging and

output power paths from being active simultaneously, discussed later in the Charging Interface and Relay Control subsection.

Electrical Design and Analysis

The Altium electrical design schematic was divided into functional circuits spanning four schematic sheets titled Buck_Converter, Power_Charging, MSP430_Controller and Top_Level_Connections. Together the sheets are responsible for power distribution, charging control, voltage conversion, system monitoring, and embedded control. The following subsections examine these circuits and component choices in flow of power and measurement signals in the system, beginning with the battery input. The complete electrical schematics are provided in Appendix A.

Battery Input, Protection, and Power Distribution

The system is powered by a 3S 2200 mAh lithium-polymer battery with a nominal voltage of 11.1 V. Each cell operates between 3.0 V and 4.2 V, establishing the battery-voltage range used throughout the design:

$$V_{BAT,min} = 3 \times 3.0 \text{ V} = 9.0 \text{ V}$$

$$V_{BAT,max} = 3 \times 4.2 \text{ V} = 12.6 \text{ V}$$

At its nominal voltage, the battery stores approximately 24.4 Wh of energy:

$$E_{BAT} = (11.1 \text{ V})(2.2 \text{ Ah}) = 24.4 \text{ Wh}$$

The XT60 connector carries the main discharge current used by the buck converter and system logic.

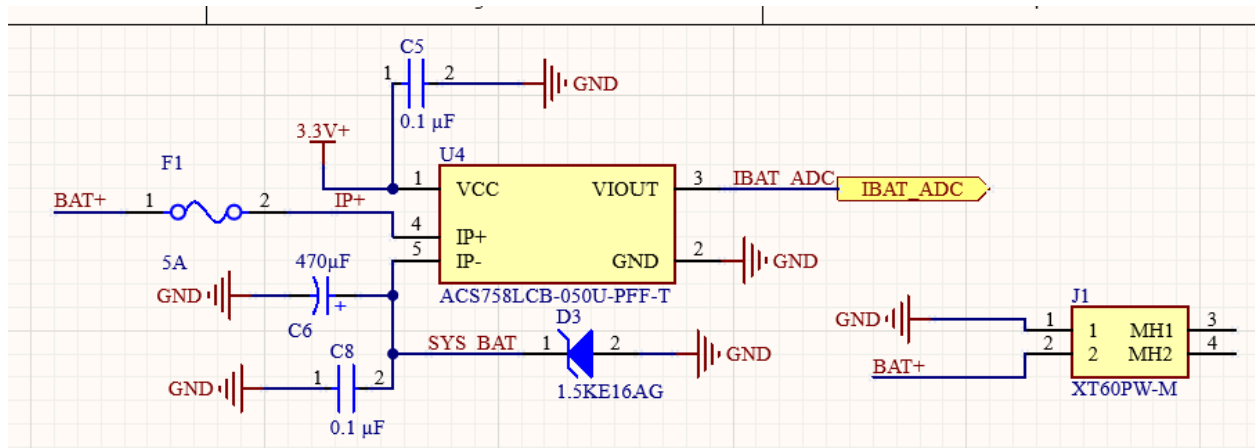


Figure 2: Battery input, protection, current sensing, and SYS_BAT output

The positive discharge path includes a replaceable automotive blade fuse and an ACS758 Hall-effect current sensor (U4) before power is distributed through the SYS_BAT rail (figure 2).

The fuse is intended to interrupt sustained overcurrent and the polarized XT60 connector reduces the possibility of accidentally reversing the battery connection. The maximum intended output power was 22.5 W at the 9 V, 2.5 A output setting. Using an estimated converter efficiency of 85% at the minimum battery voltage, the expected battery current was calculated as:

$$I_{_BAT} \approx P_{_OUT} / (V_{_BAT} \times \eta)$$

Substituting the values:

$$I_{_BAT} \approx 22.5 \text{ W} / (9.0 \text{ V} \times 0.85) \approx 2.93 \text{ A}$$

This established value was used to help select the 5A fuse value to protect against severe faults. A 1.5KE15A unidirectional TVS diode is connected between SYS_BAT and ground to suppress short-duration positive-voltage transients to protect against brief voltage excursions. The ACS758 measures the total current drawn through the battery's discharge path.

After the protection and sensing components, SYS_BAT supplies the buck converter and the 3.3 V logic regulator. The battery negative forms the common return and connects to the PCB ground plane to provide low-impedance distribution between the conversion, sensing, and control circuits.

3.3 V Logic Supply and Operating-Mode Selection

A double-pole, center-off switch controls both logic power and operating-mode selection (S2) (Figure 3). One pole connects SYS_BAT to LOGIC_PWR when either CHARGE or OUTPUT is selected. In the center OFF position, this connection is opened, removing power from the control electronics.

The LOGIC_PWR input is converted to 3.3 V by a non-isolated switching regulator (IC1). Its 7–36 V input range covers the battery’s full 9.0–12.6 V operating range. The 3.3V+ rail powers the MSP430, ADS1115 modules, OLED display, relay coils, pull-up resistors, and supporting measurement circuits. A switching regulator was chosen over a linear regulator to reduce power dissipation. The C9 and C10 capacitors provide local input filtering, while C11 and C12 provide output filtering and high-frequency bypassing to help maintain a stable logic rail as the system switches operating states.

These signals are used by the MSP430 to determine the requested operating mode. CHARGE_ALLOWED is also used to activate the charging relay driver described in the following subsection. BUCK_ALLOWED indicates that OUTPUT mode has been selected, but

does not immediately activate the buck converter output. The converter must still be enabled through an additional MSP430 output-enable control signal before the switching begins.

Charging Interface and Relay Control

The charging interface allows the installed battery to be charged through an external DC port without removing it from the enclosure. The custom PCB controls when the battery charging board is connected to the battery. An ACS758 Hall-effect sensor (U5) (figure 3) measures current in the charging path, and its analog output is acquired by an ADS1115 for display on the OLED. The sensor's scaling and current-conversion calculations are presented later in the current-measurement subsection.

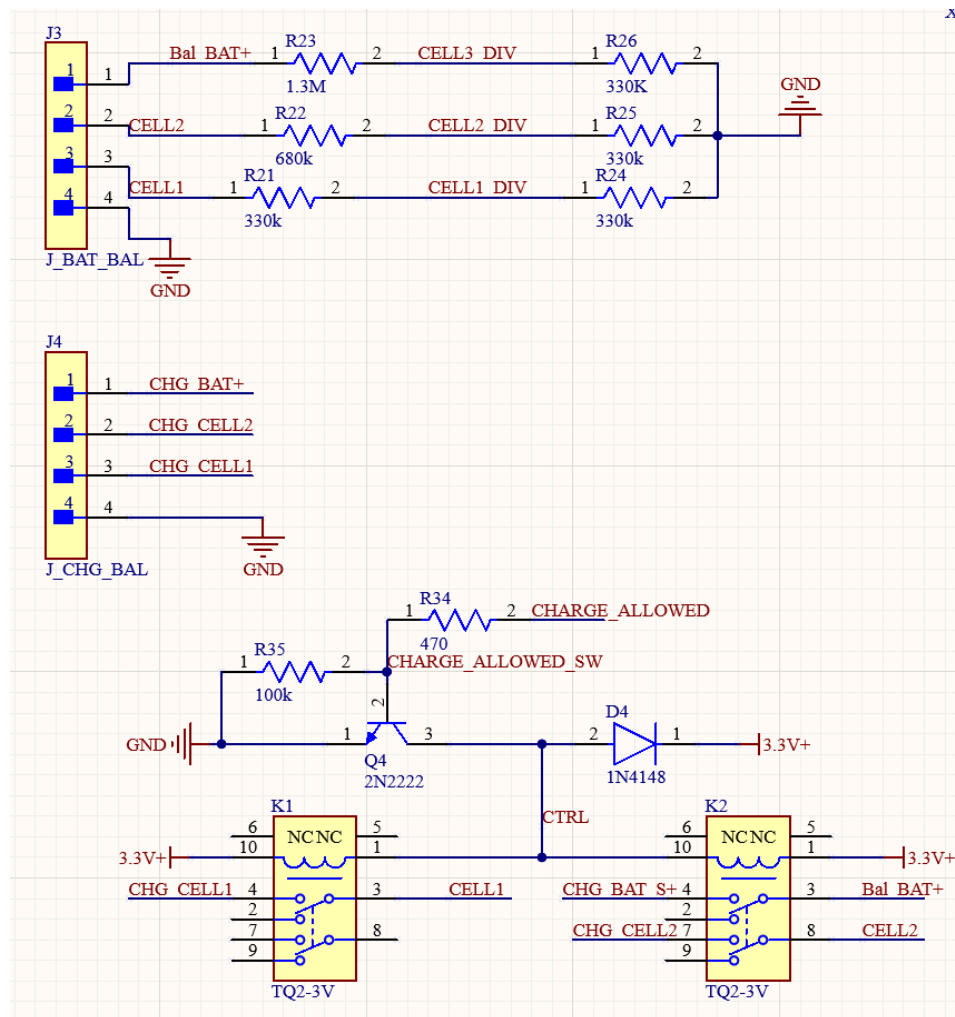


Figure 4: Charging interface and relay-control

Selecting CHARGE powers the 3.3V logic supply and creates the hardware generated CHARGE_ALLOWED signal which is applied to a NPN transistor (Q4) which acts as a low-side driver for a pair of TQ2-3V dual-pole relays (K1-2) (figure 4). This design allows the relay current to be controlled without passing the full coil current through the mode-selection signal.

When charge allowed is live, Q4 energizes the coils and the relay contact pairs switch the non-ground conductors of the battery's balancing connection. The normally open contacts keep the charger disconnected whenever the relay coils are unpowered.

Diode D4 is connected across the relay coils as a flyback diode. It is reverse-biased while the relays are energized but conducts when Q4 turns off, providing a path for the coils' stored magnetic energy and limiting the resulting voltage transient.

In OUTPUT or OFF mode, CHARGE_ALLOWED is held low, Q4 remains off, and the charging connections remain open. The same mode switch S2 holds BUCK_ALLOWED low during charging, establishing a hardware basis for preventing the charger and buck converter from operating simultaneously.

Buck-Converter Design

The power-conversion stage is a custom asynchronous buck converter designed to produce selectable 3.3 V, 5 V, and 9 V outputs from the 3S battery. The circuit combines a TL494 PWM controller, an IRS2110 high-side gate driver, an N-channel power MOSFET, a Schottky catch diode, a 68 μ H inductor, output capacitors, and a rotary-switch-selected feedback network. The following subsections describe the design and selection of each portion of the converter.

Converter Requirements and Duty-Cycle Range

For an ideal buck converter operating in continuous conduction mode, output voltage is related to input voltage and duty cycle by:

$$D = V(\text{OUT}) / V(\text{IN})$$

Using the battery's 9.0–12.6 V operating range, the ideal duty cycles for the three output settings are:

Selected output	At 12.6V	At 11.1V	At 9.0V
3.3V	26.2%	29.7%	36.7%
5.0V	39.7%	45.0%	55.6%
9.0V	71.4%	81.1%	100%

Actual operation requires additional duty cycle to compensate for losses in the MOSFET, catch diode, inductor, and PCB conductors.

TL494 PWM Control and Oscillator

The TL494 provides the fixed-frequency pulse-width-modulation signal used to regulate the buck converter. Its internal oscillator generates a waveform that is compared with the control-loop signal to vary the PWM duty cycle. The controller's two output transistors are

configured for single-ended operation because the converter requires one PWM signal for the IRS2110 high-side gate driver.

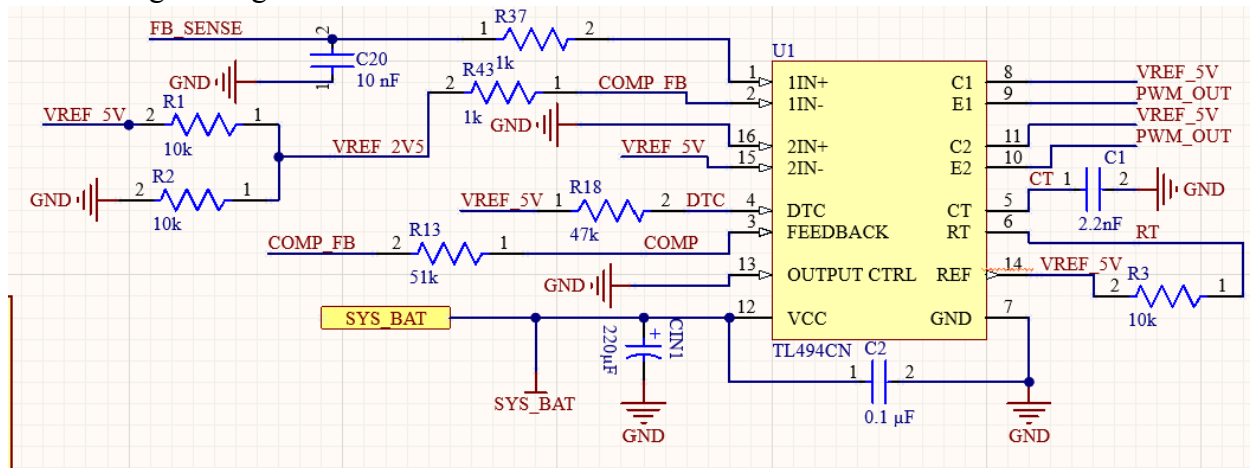


Figure 5:TL494 PWM-control

The oscillator frequency is established by timing resistor (R3) and timing capacitor (C1), connected to the RT and CT pins (figure 5). For single-ended operation, the approximate oscillator and switching frequency is:

$$f_{SW} \approx f_{OSC} \approx 1 / (R3 \times C1)$$

Using the final component values from the schematic:

$$f_{SW} \approx 1 / (10k \times 2.2nF)$$

$$f_{SW} \approx 45.5\text{kHz}$$

Because the TL494 is configured for single-ended operation, the output switching frequency is equal to the oscillator frequency.

The TL494 produces a stable internal 5 V reference, but this cannot be used directly as the comparison voltage because the lowest regulated output is 3.3 V. Therefore, the 5 V reference is divided by equal 10 k Ω resistors:

Resistor divider equation:

$$V_{out} = V_{in} \times (R2 / (R1 + R2))$$

Using component values:

$$V_{\text{out}} = 5 \text{ V} \times (10 \text{ k}\Omega / (10 \text{ k}\Omega + 10 \text{ k}\Omega))$$

$$V_{out} = 2.5V$$

Using 2.5 V provides a comparison level below all three outputs 3.3 V, 5 V, and 9 V so each setting can be sensed using resistor dividers inside a passive output feedback network.

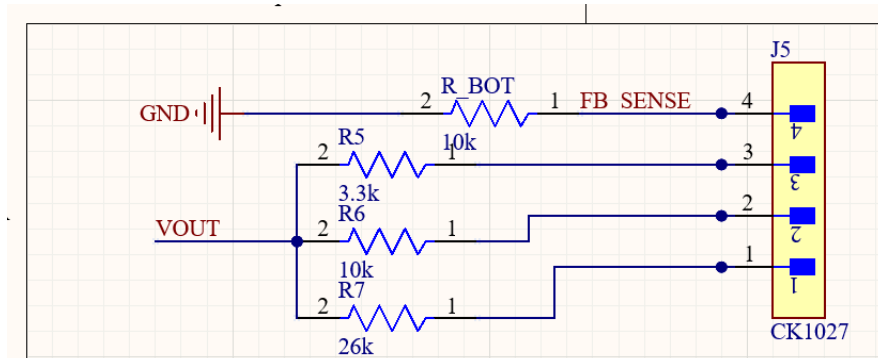


Figure 6: TL494 resistor feedback network

Figure 6 shows the resistor feedback network used to select the regulated output voltage. The 4 pole 3 position rotary switch selects a different resistance ratio for each of the 3.3 V, 5 V, and 9 V settings. The TL494 feedback pin receives a divided version of the output voltage with the lower resistor R_{BOT} fixed at 10k Ω :

$$V_{FB} = V_{OUT} \times (R_L / (R_U + R_{BOT}))$$

Each switch position is designed so that $V_{FB} \approx 2.5$ V when the output voltage is correct.

Rearranging the divider equation gives the required resistor ratio:

$$R_U / 10k\Omega = (V_{OUT} / 2.5 \text{ V}) - 1$$

Resulting in the desired values values:

(3.32k and 26.1k resistors used in final design due to better availability)

3.33V, 5V and 9V

Changing the selected resistance changes the output voltage required to produce the same 2.5 V feedback level. The rotary switch J5 therefore selects the output voltage directly through hardware rather than through the MSP430. Second and third poles on the rotary switch are used to communicate which voltage has been selected to MSP430 and illuminate a corresponding led further explained in the MSP430 and Peripheral Hardware subsection.

The FB_SENSE signal resulting from the feedback network passes through R37, while C20 filters high-frequency switching noise before the signal reaches the non-inverting input of the first TL494 error amplifier (Figure 5). The 2.5 V reference is applied to the amplifier's inverting input. If the output voltage fluctuates the amplifier changes the PWM duty cycle accordingly to maintain the selected output as the battery voltage or load current changes. Only one TL494 error amplifier is required for voltage regulation. The other is forced low by connecting its non-inverting input, 2IN+, to ground and its inverting input, 2IN-, to the 5 V reference to prevent it from interfering with PWM control.

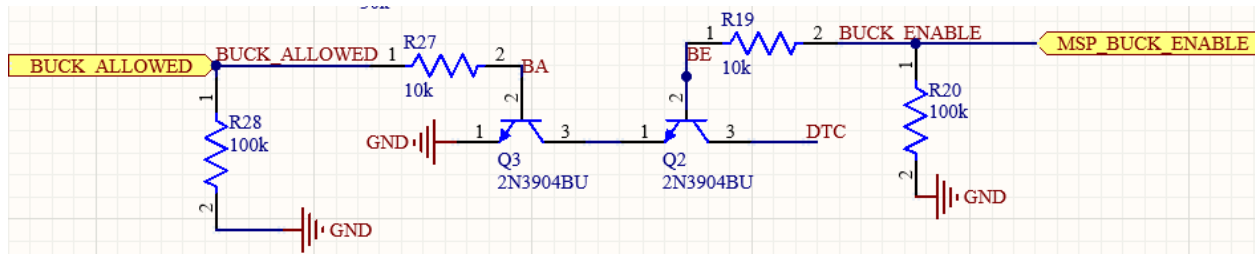


Figure 7: PWM enable circuit

The dead-time-control input provides the converter's default-off function. R18 pulls DTC toward 5 V when the control transistor is inactive, inhibiting PWM generation (figure 5). After OUTPUT mode is selected and BUCK_ALLOWED enables the first transistor the MSP430-controlled BUCK_ENABLE transistor pulls DTC low and permits PWM (figure 7). Operation. This design keeps the converter disabled by default if the MSP430 is unpowered or initializing, preventing the buck converter from starting automatically when the operating-mode switch is moved to OUTPUT.

IRS2110 gate driver and MOSFET

The two TL494 output transistors are connected in parallel to produce a single PWM signal for the IRS2110 gate driver. Their collectors are connected to the 5 V reference, their emitters are joined at PWM_OUT, and the output-control input is grounded to select single-ended operation.

The ground-referenced PWM signal produced by the TL494 cannot directly drive the high-side N-channel MOSFET because the MOSFET source rises toward the battery voltage when it turns on. The gate must remain several volts above the source to maintain the required gate-to-source voltage. Therefore, an IRS2110 high- and low-side gate driver was chosen to translate PWM_OUT into a floating high-side gate-drive signal (U2) (figure 8).

As shown in Figure 8 PWM_OUT is applied to the IRS2110 high-side input, while lowside remains inactive. The drivers' HO output controls the gate of the STP36NF06L MOSFET through a series gate resistor which limits peak gate current and reduces ringing caused by inductance and capacitance in the switching path. The IRS2110 requires a floating supply to drive the high-side MOSFET. This floating supply voltage is maintained by the bootstrap diode Cbot between the switch node and the bootstrap supply node which charges through D2 when the MOSFET is off using SYS_BAT. When the MOSFET is turned on the Cbot's voltage is used to "boost" HO above the Mosfet source.

$$V \setminus G \approx V \setminus S + V \setminus BS$$

The gate-to-source drive voltage therefore remains approximately equal to the voltage stored across the bootstrap capacitor even as the source rises toward SYS BAT. This allows for the

N-channel Mosfet to remain fully driven while operating as a high side switch. The gate-drive supply range begins at 10 V, while the battery can fall to approximately 9.0 V. Because the bootstrap capacitor must recharge during off time the circuit is limited to < 100% duty-cycle, preventing optimal output of 9V at very low battery capacity.

The STP36NF06L is a 60 V N-channel MOSFET rated for 30 A, with a maximum on-state resistance of 40 mΩ at a 10 V gate drive and 50 mΩ at a 5 V gate drive. Its 60 V drain-to-source rating provide a comfortable margin above the maximum 12.6 V battery voltage:

$$\text{Voltage margin} = 60 \text{ V} / 12.6 \text{ V} \approx 4.8$$

Using the conservative 50 mΩ on-resistance and approximating the MOSFET current as 2.5 A, the maximum first-order conduction-loss estimate is 0.31 watts of heat meaning the device should run cool and is well suited for this application. (omits switching and gate-charge losses)

Given Voltage across MOSFET when ON: $V = I \times R_{\text{DS(on)}}$

Power dissipated:

$$P = V \times I = I \times (I \times R_{\text{DS(on)}}) = I^2 \times R_{\text{DS(on)}}$$

$$P_{\text{CONDL}} \approx I_{\text{MOSFET}}^2 \times R_{\text{DS(on)}} \times D$$

Assuming unity duty cycle ($D = 1$):

$$P_{\text{CONDL}} \leq (2.5 \text{ A})^2 \times 0.050 \Omega \times 1$$

$$P_{\text{CONDL}} = 0.31 \text{ W}$$

Catch Diode, Inductor, and Output Filtering

The MOSFET, catch diode, inductor, and output capacitors form the buck converter's main power stage, shown in Figure 9. When the MOSFET is on, current flows from SYS_BAT through the MOSFET and inductor to the output. During this interval, the inductor stores energy while simultaneously supplying the load and charging the output capacitors. When the MOSFET turns off, the inductor current cannot change instantaneously. As the magnetic field collapses, the inductor continues driving current into the load. This current returns through the catch diode, which is oriented with its cathode at SW_NODE and its anode at ground. The diode is reverse-biased during the MOSFET on-time and forward-biased during the off-time, providing a continuous current path for the inductor. The STPS5L60 Schottky diode was chosen for its low forward voltage drop, fast switching and ability to withstand more than the 12.6V maximum battery voltage while reverse-biased and conduct peak inductor current during off-time.

The converter uses a 68 μH inductor. Inductor ripple current can be approximated by:

$$\Delta I_L = (V_{\text{IN}} - V_{\text{OUT}}) \times D / (L \times f_{\text{SW}})$$

The largest ripple occurs near the 5 V output setting with a fully charged 12.6 V battery. Using

$$\begin{aligned}
 D &= 5 / 12.6 = 0.397, \\
 L &= 68 \mu\text{H}, \\
 f_{\text{SW}} &= 45.5 \text{ kHz:} \\
 \Delta I_L &= (12.6 - 5) \times 0.397 / (68 \times 10^{-6} \times 45.5 \times 10^3) \\
 \Delta I_L &\approx 0.98 \text{ A (peak-to-peak)}
 \end{aligned}$$

Ripple swings half above and half below the average current. At the maximum intended 2.5 A output current, the approximate peak and minimum inductor currents become:

$$I_{L,\text{PEAK}} = I_{\text{OUT}} + (\Delta I_L / 2)$$

$$I_{L,\text{PEAK}} = 2.5 \text{ A} + (0.98 \text{ A} / 2)$$

$$I_{L,\text{MIN}} = 2.5 \text{ A} - (0.98 \text{ A} / 2)$$

$$I_{L,\text{PEAK}} \approx 2.99 \text{ A}$$

$$I_{L,\text{MIN}} \approx 2.01 \text{ A}$$

Because the minimum current remains above zero, the converter operates in continuous-conduction mode at the maximum load under this condition. The approximate RMS inductor current (thermal stress) is:

Triangular ripple current equation:

$$I_{L,\text{RMS}} = \sqrt{I_{\text{OUT}}^2 + (\Delta I_L^2 / 12)}$$

$$I_{L,\text{RMS}} = \sqrt{(2.5 \text{ A})^2 + (0.98 \text{ A})^2 / 12}$$

$$I_{L,\text{RMS}} \approx 2.52 \text{ A}$$

The inductor should have a saturation-current rating above 3.0 A and an RMS-current rating above 2.52 A, with additional margin for transients, tolerances, and heating.

The output capacitors reduce voltage ripple and support the load during transients. The bulk capacitor handles lower-frequency changes, while the ceramic capacitor filters high-frequency switching noise. Similarly, the input capacitors supply the MOSFET's pulsed current locally and reduce disturbances on the battery and logic-supply connections.

LTspice Simulation and Design Verification

Following the analytical calculations, an idealized LTspice model was created to verify the expected behavior of the buck-converter power stage. As shown in Figure 9, the simulation used the final 45.5 kHz switching frequency, 68 μH inductor, and 220 μF and 22 μF output capacitors. The input was set to the fully charged battery voltage of 12.6 V, and the load was adjusted to represent the target 2.5 A output current.

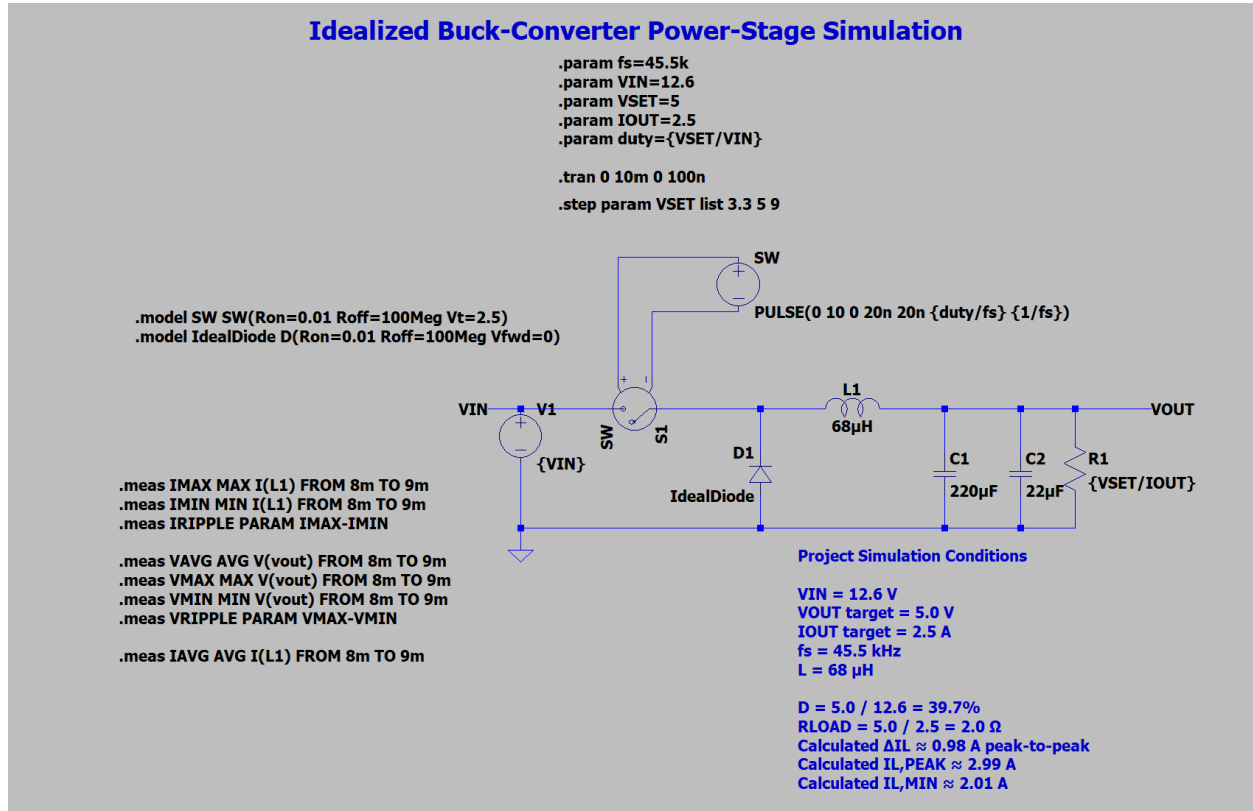


Figure 9: LTspice model

The simulation used the final design values and the load resistance was adjusted for each output setting so that the simulated load current remained at 2.5 A:

Output setting	Ideal duty cycle	Load resistance
3.3 V	26.2%	1.32 Ω
5.0 V	39.7%	2.00 Ω
9.0 V	71.4%	3.60 Ω

The three operating conditions were evaluated using an LTspice parameter sweep. As shown in Figure 10, the simulated outputs settled near 3.28 V, 4.98 V, and 8.98 V. The approximately 0.02 V difference from each target resulted primarily from the 0.01 Ω on-state resistance included in the ideal switch model:

$$V_{\text{DROP}} = I_{\text{OUT}} \times R_{\text{ON}}$$

$$V_{\text{DROP}} = 2.5 \text{ A} \times 0.01 \text{ } \Omega = 0.025 \text{ V}$$

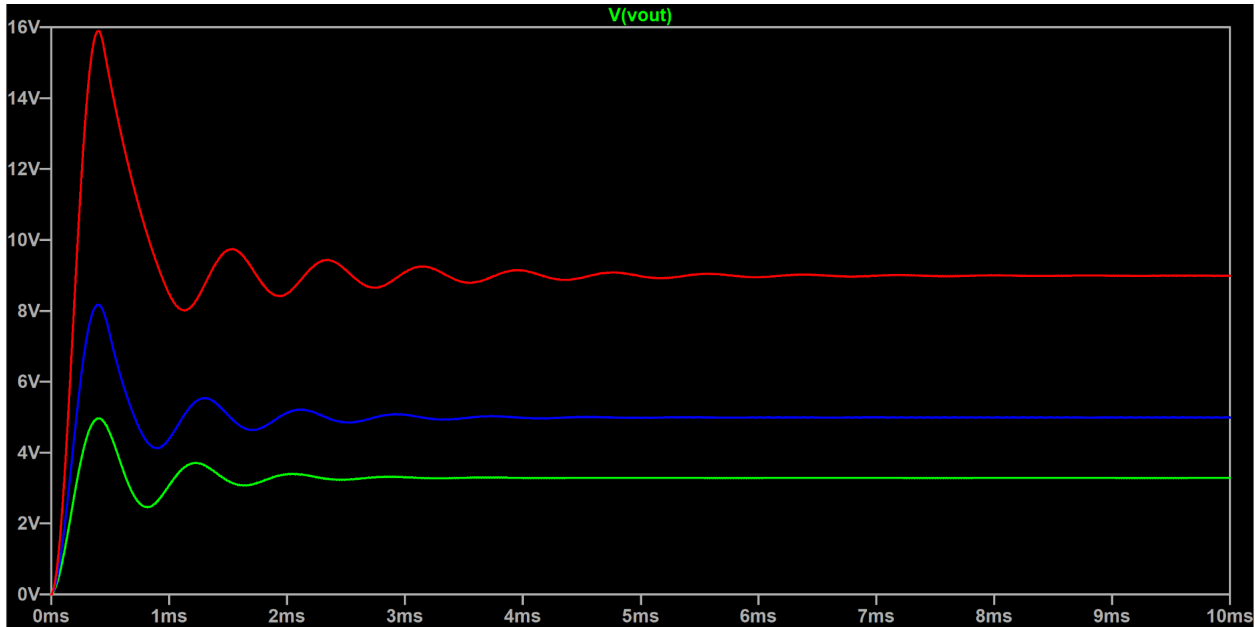


Figure 10: LTspice simulation

As shown in Figure 11, the simulated outputs settled near 3.28 V, 4.98 V, and 8.98 V. Each result was within 0.02 V of its intended value due to the switch model. These results supported the previously calculated relationship between input voltage, duty cycle, and output voltage. Steady-state measurements were taken between 8 ms and 9 ms so that the initial LC-filter startup response did not affect the results.

The simulation showed some startup oscillation, particularly at the 9 V setting, but it decayed as the output approached steady state. Because the model used an idealized switch and diode, it did not reproduce the complete behavior of the TL494 controller, IRS2110 gate driver, DMT6009LCT MOSFET, STPS5L60 catch diode, or physical PCB parasitics. Overall, the simulation confirmed the required duty-cycle range, appropriate operation of the 68 μH inductor and 242 μF output filter, and delivery of approximately 2.5 A at all three selected output voltages, while closed-loop regulation and actual component behavior were evaluated during hardware testing.

Output-Voltage Measurement

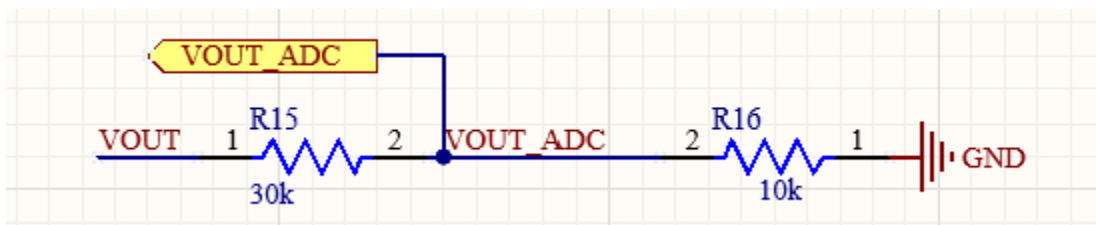


Figure 10: LTspice simulation

The buck converter output voltage is measured using the resistor divider shown in figure 12. R15 and R16 reduce the maximum 9 V output to a level that can be safely measured by the ADS1115.

The divider output is:

$$V_OUT_ADC = V_OUT \times [R16 / (R15 + R16)]$$

Using R15 = 30 k Ω and R16 = 10 k Ω :

$$V_OUT_ADC = V_OUT \times [10\text{ k}\Omega / (30\text{ k}\Omega + 10\text{ k}\Omega)]$$

The expected ADC voltages are therefore 0.825 V, 1.25 V, and 2.25 V for the 3.3 V, 5 V, and 9 V output settings, respectively. The 2.25 V value at 9V allows for potential higher voltages to be safely read during startup. The MSP430 reads this signal through the ADS1115 and reconstructs the output voltage using:

$$V_OUT = V_OUT_ADC \times 4$$

This measurement is used for monitoring and display. It is separate from the TL494 feedback network responsible for regulating the converter output.

Current-Measurement Circuits

Three current-measurement channels are included to monitor current at different points in the system. A shunt-resistor circuit measures the regulated buck-converter output, while two ACS Hall-effect sensors measure battery-pack and charging current.

Buck Output-Current Measurement

The buck converter's output current is measured using the low-side shunt circuit shown in Figure 11. The shunt resistor is located between the negative output terminal and system ground, forcing the load-return current to pass through it.

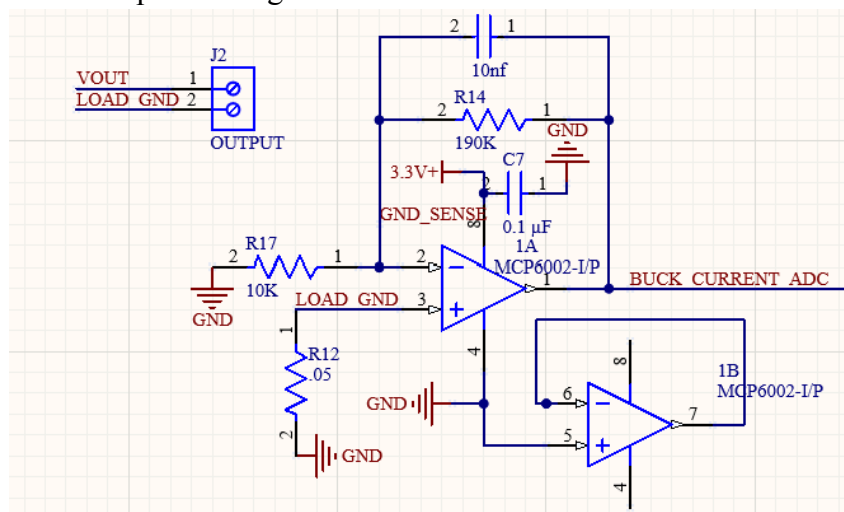


Figure 10: Buck converter current sense circuit

The voltage developed across the shunt is:

$$V_SHUNT = I_OUT \times R_SHUNT$$

Because this voltage is relatively small, one channel of the MCP6002 operational amplifier increases it before it is measured by an ADS1115 analog-to-digital converter. The relationship between output current and the amplified signal is:

$$V_ADC = I_OUT \times R_SHUNT \times G$$

Therefore, the output current can be calculated as:

$$I_OUT = V_ADC / (R_SHUNT \times G)$$

The 10nf capacitor in parallel with the 190kΩ feedback resistor together creates a low-pass filter with a cutoff around 84Hz.

$$f_c = 1 / (2\pi RC)$$

$$f_c = 1 / (2\pi \times 190000 \Omega \times 10 \times 10^{-9} F)$$

$$f_c \approx 84 \text{ Hz}$$

This value was chosen to filter out 45.5HZ switching ripple and preserve slow changes in load current which are sent to the ADC1115.

Battery and Charging-Current Measurement

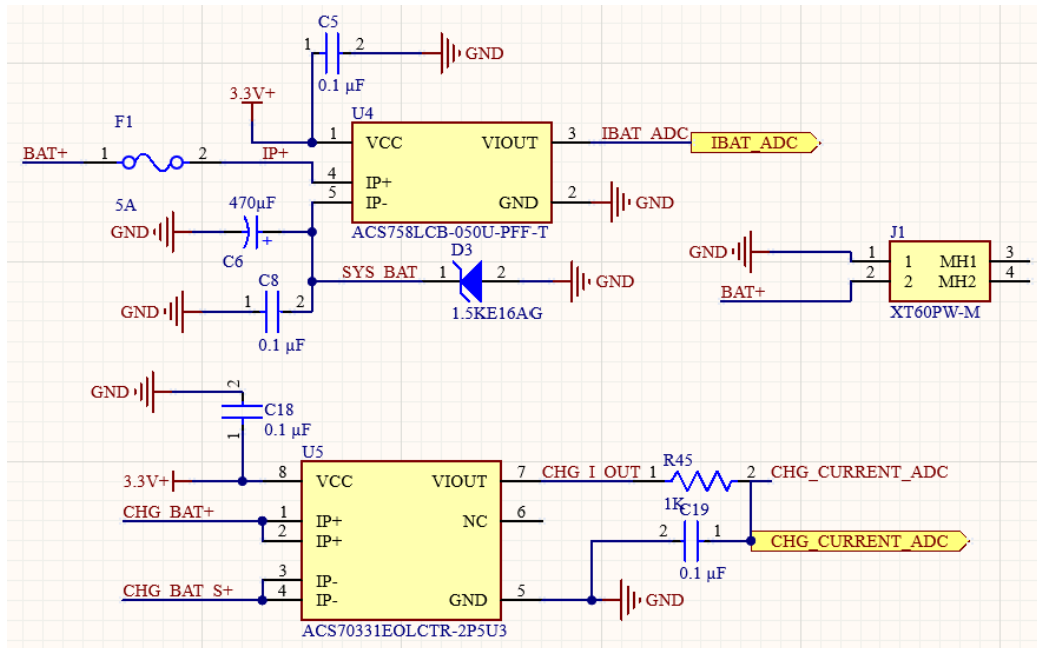


Figure 12: Buck converter current sense circuit

An ACS758LCB-050U-PFF-T Hall-effect sensor (U4) is connected between BAT+ and SYS_BAT, following the 5 A fuse. Current flowing from the battery to the system passes through the sensor's low-resistance internal conductor. The resulting IBAT_ADC signal represents the total battery current consumed by the buck converter and supporting electronics. At the estimated maximum battery current of 2.94 A, the expected sensor output is approximately 0.512 V. Because the 50 A range is much greater than the normal system current, the sensor provides substantial current capacity but limited sensitivity at low currents further explained in the Battery-Current Measurement Limitation subsection.

Charging current is measured using an ACS70331EOLCTR-2P5U3 GMR current sensor (U5) connected between CHG_BAT+ and CHG_BAT_S+. This unidirectional sensor was selected for its 0–2.5 A measurement range and higher sensitivity.

The sensor has a nominal zero-current output of 0.25 V and a sensitivity of 0.8 V/A. Charging current is therefore calculated using:

$$I_CHG = (V_CHG_CURRENT_ADC - 0.25 \text{ V}) / 0.8 \text{ V/A}$$

R45 and C19 form a low-pass filter that reduces high-frequency noise before the signal reaches the ADS1115.

Temperature Measurement

Battery temperature is measured using a 10k Ω negative-temperature-coefficient thermistor. at 25°C and forms a voltage divider with the 10 k Ω fixed resistor shown in Figure 13.

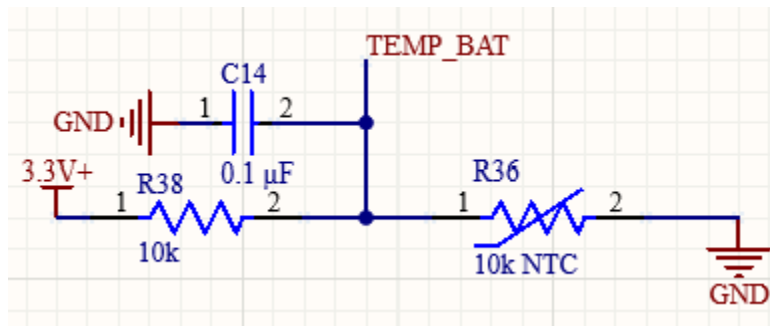


Figure 13: Battery-temperature measurement circuit

In the final circuit, the NTC thermistor is connected between ground and TEMP_BAT, while the fixed resistor is connected between TEMP_BAT and 3.3V. At 25°C, the two resistances are approximately equal, producing:

$$V_TEMP = 3.3 \text{ V} \times [10 \text{ k}\Omega / (10 \text{ k}\Omega + 10 \text{ k}\Omega)]$$

$$V_TEMP \approx 1.65 \text{ V}$$

As the battery temperature increases, the thermistor resistance decreases, causing the TEMP_BAT voltage to decrease. The thermistor resistance is calculated from the measured voltage using:

$$R_{_NTC} = 10 \text{ k}\Omega \times [V_{_TEMP} / (3.3 \text{ V} - V_{_TEMP})]$$

The ADS1115 digitizes the signal, and the MSP430 converts the calculated resistance into temperature using the thermistor's resistance-temperature relationship. C14, a 0.1 μF capacitor, filters electrical noise at the divider output and stabilizes the temperature measurement.

MSP430 and Peripheral Hardware

An MSP430G2553 microcontroller provides system management, measurement processing and user-interface control. The microcontroller operates from the regulated 3.3V logic supply and is programmed using an MSP430 LaunchPad (figure 14).

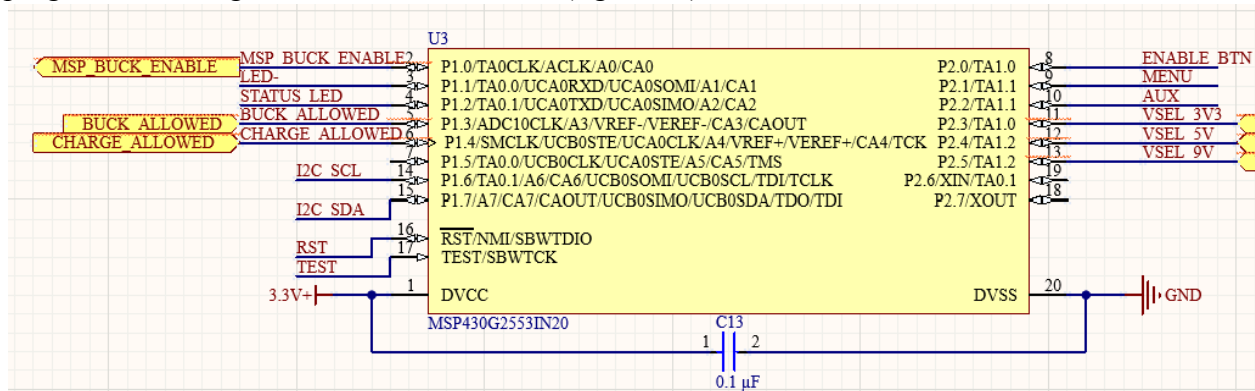


Figure 14: MSP430 connections

The MSP430 monitors the front-panel controls to determine the requested operating mode and whether the buck converter has been enabled. It generates the MSP_BUCK_Enable control signal used to activate the second transistor in the DTC buck enable circuit.

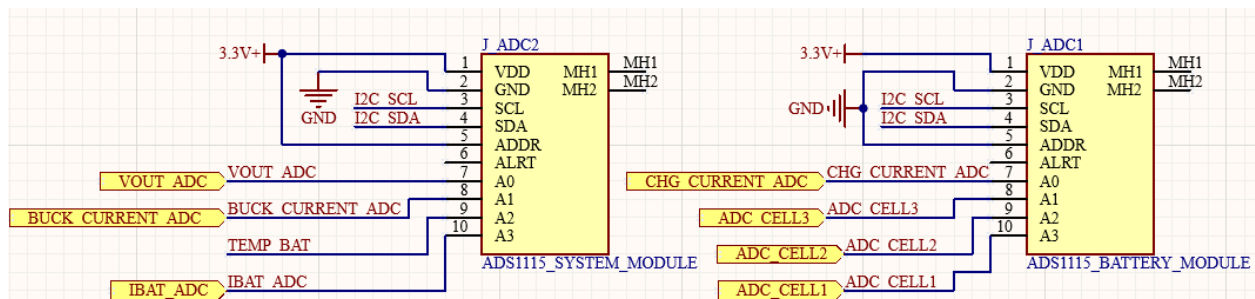


Figure 15: ADS1115 analog-to-digital converter modules

Two ADS1115 analog-to-digital converter modules provide the analog-measurement channels required by the system. The converters are assigned different I²C addresses using 3.3V+ and GND so they can share the same bus. They digitize the battery cell voltages, buck output voltage, battery current, charging current, buck output current and temperature signals. Their higher resolution improves measurement of the relatively small current-sensor outputs compared with using only the MSP430's internal ADC. The modules also free up pins on the MSP430 for other applications.

A 2.42-inch SSD1309 OLED display is connected to the same I²C bus. The display presents the selected operating mode, battery condition, output voltage, current, and temperature.

Additional connections are provided for the illuminated enable pushbutton rotary switch and status LEDs. The three-position rotary switch selects the 3.3 V, 5 V, or 9 V output setting. The switch contains four electrically independent poles that move together when the knob is rotated, with three poles used in this design. The first pole selects the corresponding resistor path in the TL494 feedback network, establishing the regulated output voltage. The second pole communicates the selected position to the MSP430 so the firmware can display the correct voltage setting. The third pole activates the indicator status LEDs corresponding to the selected voltage. These inputs and outputs allow the firmware to respond to user commands and provide visual feedback without directly carrying power-stage current.

PCB Architecture and Component Placement

The completed circuit was implemented on a 180 mm $\times$ 90 mm four-layer PCB. As shown in Figures 16 & 17, the PCB was divided into five primary functional regions. Organizing the components by function simplified routing and helped separate the switching power stage from the sensitive analog and digital circuitry. A four-layer structure was selected because the design combines a switching converter, battery and charging paths, low-level analog measurements, and digital communication on one board.

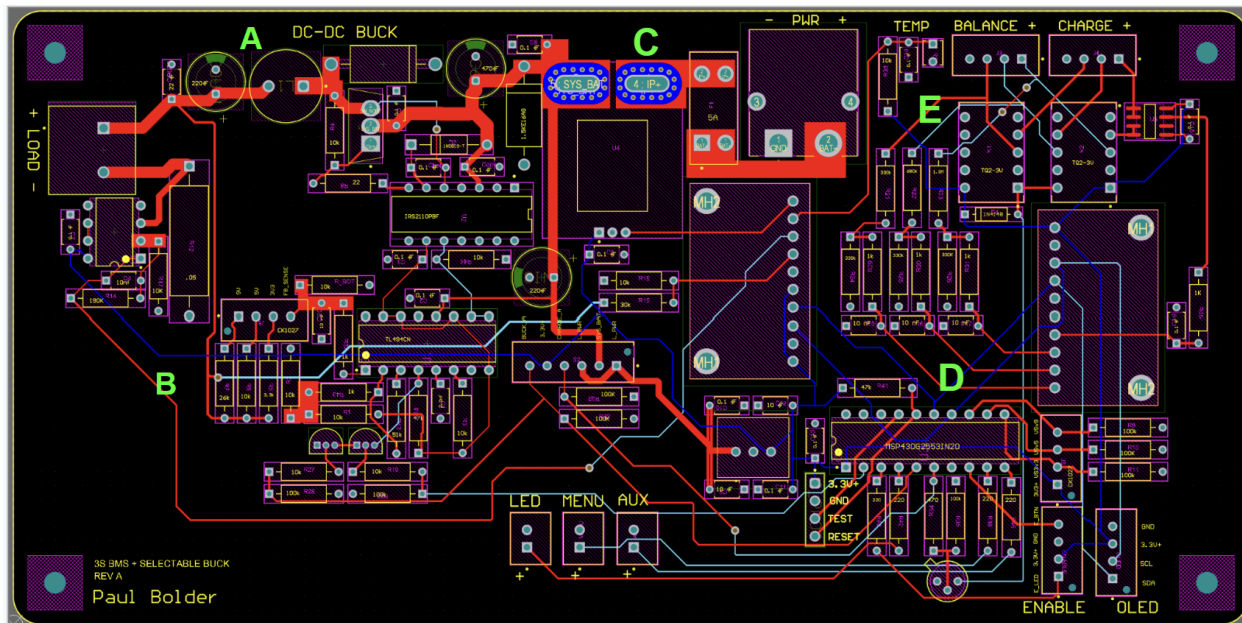


Figure 16: Altium PCB 2D Layout

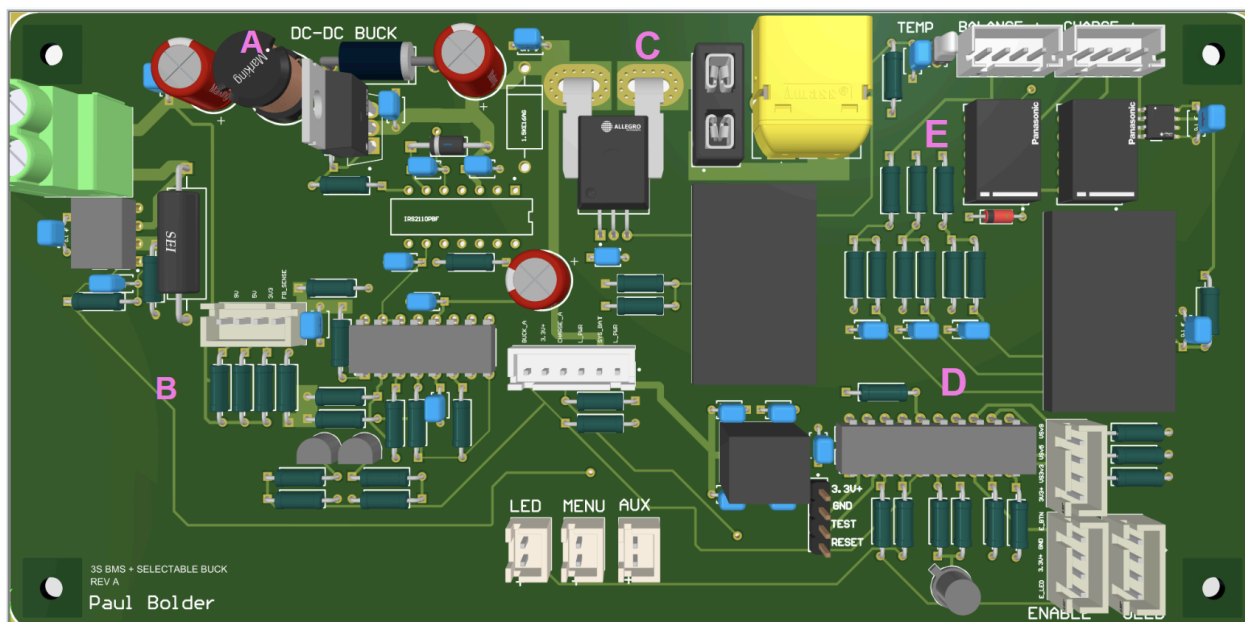


Figure 17: Altium PCB 3D Layout

A-Power Stage

Region A contains the buck-converter power stage, including the input and output capacitors, MOSFET, catch diode, inductor, and high-current output connection. These components were grouped closely to minimize the switching-current loop and reduce parasitic resistance and inductance.

B-PWM Control

Region B contains the buck-converter control and feedback circuitry which includes the TL494 PWM controller, IRS2110 gate driver, selectable feedback network, buck current sense opamp circuit, and buck enable DTC transistor circuit. Placing this circuitry near the power stage kept the gate-drive and feedback paths relatively short.

C-Power Input

Region C contains the battery power-input and distribution circuitry. The battery connection, 5 A fuse, ACS758 battery-current sensor, and associated filtering components are located here. This design allows the battery current to pass through the fuse and current sensor before reaching the rest of the system.

D-Logic Circuitry

Region D contains the MSP430G2553, ADS1115 converter modules, programming connections, 3.3V switching regulator and user-interface circuitry. Locating the digital control and measurement hardware away from the buck-converter switching region reduces the possibility of switching noise affecting the analog measurements or I²C communication.

E-Charging

Region E contains the battery-charging and balance-connection circuitry. The charging-current sensor, electromechanical relays, balance connector, three battery cell voltage dividers and charging connector are grouped together to keep the switched charging paths short and simplify connections to the external charger.

The board connectors were positioned primarily around the PCB perimeter based on the intended location of connecting peripherals inside the device enclosure to simplify wiring.

Firmware Design and System Operation

Architecture and Initialization

The MSP430G2553 firmware was written in C using Texas Instruments Code Composer Studio. The program was divided into separate source modules for clock configuration, GPIO control, timing, input processing, operating-mode control, I²C communication, ADS1115 measurements, safety, and OLED operation. A modular approach allowed for each subsystem to be tested separately before being implemented in the complete program.

At startup, the watchdog timer is stopped and the system clock, GPIO pins, timer functions, I²C interface, ADS1115 converters, and OLED display are initialized. The buck converter is explicitly placed in a disabled state during initialization. This prevents an unintended output from appearing while the microcontroller and peripheral hardware are starting.

After initialization, the firmware repeatedly performs the following primary operations:

1. Read the front-panel mode switch, enable button, and voltage-selection inputs.
2. Determine the requested operating mode.
3. Acquire the required measurements from the ADS1115 converters.
4. Convert the raw measurements into engineering units.
5. Evaluate output safety conditions.
6. Update the OLED display and indicator LED states.

The MSP430 supervises the converter but does not generate its PWM switching waveform.. Buck-converter regulation remains the responsibility of the TL494 and its analog feedback network.

The three-position front-panel switch requests OFF, CHARGE, or OUTPUT operation using its second pole. The firmware interprets the two mode-request signals and sets the charging and buck-enable outputs accordingly.

In OUTPUT mode, the enable pushbutton toggles the buck converter on and off. Changing operating modes clears the previous buck-enabled state, requiring the user to press the enable button again after returning to OUTPUT mode. If an invalid combination of mode-request signals is detected, the firmware disables the voltage output operation.

Measurement Acquisition and Processing

The firmware selects the required ADS1115 channels, initiates conversions, and converts the returned digital values into voltages. Voltage-divider scale factors are applied to reconstruct the actual battery and buck-output voltages. Because the balance connector provides cumulative cell-tap voltages, the individual cell voltages are calculated after the divider scaling has been applied:

$$V_CELL1 = V_TAP1$$

$$V_CELL2 = V_TAP2 - V_TAP1$$

$$V_CELL3 = V_PACK - V_TAP2$$

These cell values are used to monitor battery health and estimate battery life.

For temperature measurement, the firmware first calculates the NTC thermistor resistance from the measured divider voltage. It then applies the thermistor beta relationship:

thermistor Beta equation:

$$T_K = [1/T_0 + (1/B) \times \ln(R_NTC/R_0)]^{-1}$$

where R_0 is 10 k Ω at $T_0 = 25^\circ\text{C}$ and the thermistor beta value is 3380 K. The calculated temperature is converted from kelvin to degrees Celsius and then to degrees Fahrenheit for display.

The CK1027 rotary-switch position is also read by the MSP430. This input indicates whether the hardware feedback network is set for 3.3 V, 5 V, or 9 V. The firmware uses this information to display the selected voltage and control its corresponding indicator LED. Actual output-voltage is read using a voltage divider connected to the buck output.

OLED User Interface

The information shown on the 2.42-inch OLED changes according to the selected operating mode. In OUTPUT mode, the display shows the selected voltage, measured output voltage, buck-output current, battery condition, temperature, and whether the output is on or off. The illuminated enable button remains on while the buck converter is active.

In CHARGE mode, the output measurements are replaced by charging information. The display shows the status, charging current, battery cell voltages, battery level icon. If the enable button is pressed while charging, the buck converter remains disabled while LEDs and the charging indicator on the OLED screen flash to signal that output is not available.

In OFF mode no voltage is sent to the 3.3V switching regulator powering the logic circuit and all charging and output functions are disabled.

Figures 18 and 19

Safety Features

The system monitors and compares both temperature and output-current values against set limits to determine if conditions are safe for continued operation. The MSP 430 buck enable signal remains active only while OUTPUT mode is selected, the user has pressed the enable button, and safety measurements are within permitted limits.

The programmed Output-current trip threshold is 2.7 to provide a small margin above the converter's 2.5A target load. If the calculated output current reaches or exceeds this threshold

output is stopped. A separate 3A inline fuse is installed in series with the buck-converter output to protect the converter, output wiring, and connected load if the output terminals are shorted.

The battery upper temperature limit is set to 95 degrees. If the limit is exceeded the buck converter is disabled and the OLED alternates between “OUTPUT OFF” and “HIGH TEMPERATURE.” The enable-button LED flashes with the warning while the temperature remains outside the permitted range.

Mechanical Design and System Integration

Enclosure Design

The enclosure was designed in Onshape to combine the PCB, battery, charging hardware, user controls, and external connections. The enclosure was designed around the 180 mm × 90 mm main PCB while also providing space for the 3S lithium-ion battery and balance-charger module. The outer body uses rounded corners and beveled upper and lower edges and the lid includes a handle to improve portability. An 8 mm wall thickness provides mechanical rigidity and sufficient material around the larger panel openings.

The height of the enclosure (excluding the handle) was set at 4 inches to provide room for taller components and to make wiring easier. On the left side of the enclosure are openings for the Output wires and charging cable which allow the device to be used and recharged without removing the battery.

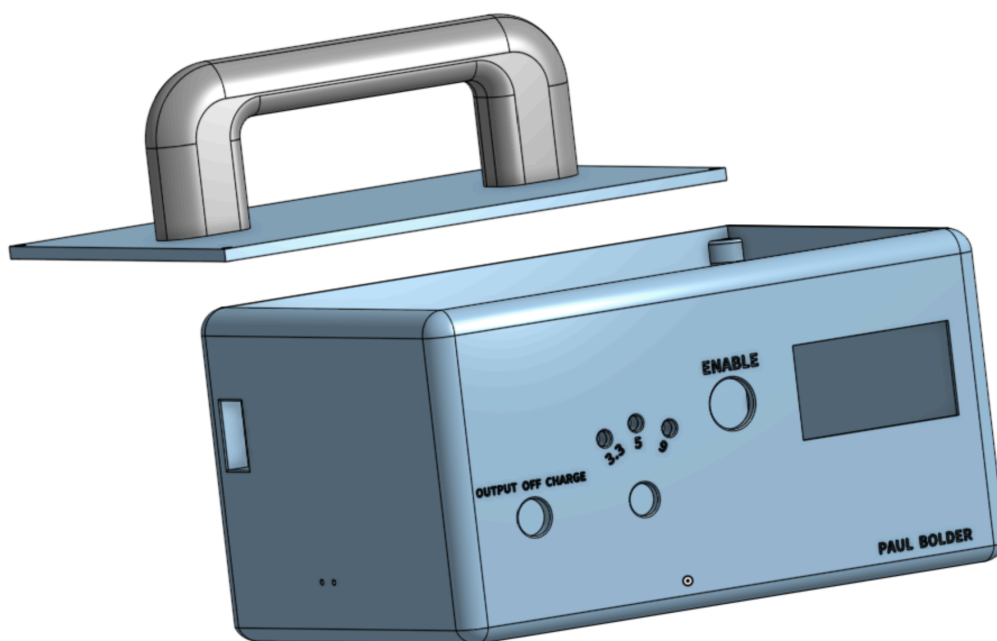


Figure 20: Enclosure 3D CAD model front

The front panel contains the controls and information required during normal operation. From left to right, the panel includes the three-position operating-mode switch, CK1027 rotary voltage

selector, illuminated enable pushbutton, and OLED display. The CK1027 rotary switch is fitted with a 24 mm knob and labeled for the available 3.3 V, 5 V, and 9 V settings. Three 5 mm indicator LEDs provide an immediate visual indication of the selected voltage. The voltage labels and LEDs are positioned around the switch in intervals of 30 degrees so that their relationship to each rotary position remains clear.

The enable button's internal LED indicates when the converter is active and is also used by the firmware to communicate charging and fault conditions.

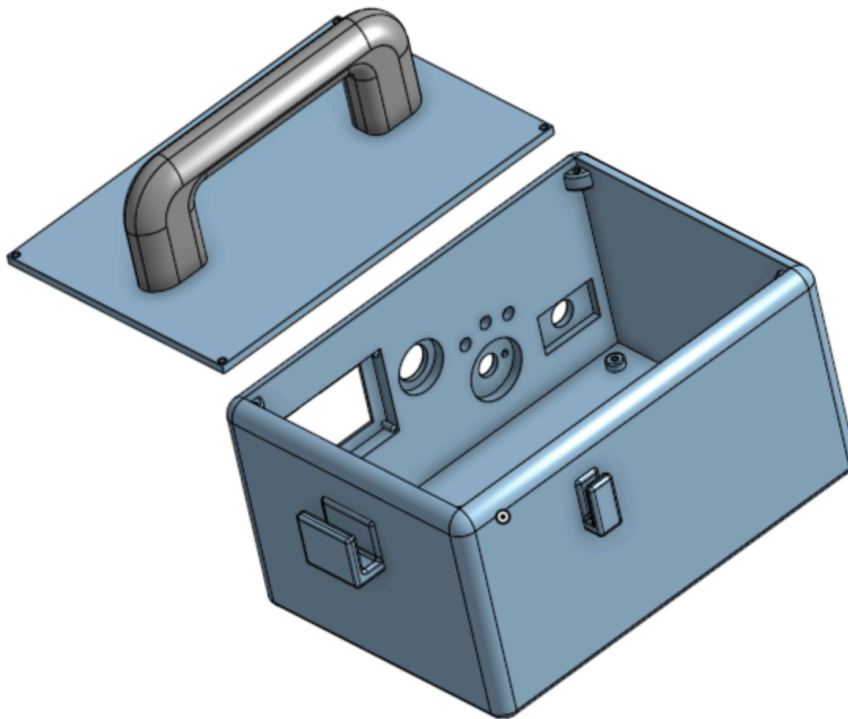


Figure 21: Enclosure 3D CAD model rear

Located on the sides of the enclosure are clips to hold the buck output cables and the charging cable when they are not in use. 4mm risers sit inside the enclosure with screw holes to hold the PCB and the charging module in place and prevent through hole components from making contact with the enclosure below.

Figure 22: Device wiring

Figure 23: Final assembly

The total model required about 18 hours to print and 600g of PTEG 3D filament.

Testing and Validation

Functional and Load Testing

Functional testing was performed after assembling the PCB, firmware, and user-interface hardware. The OFF, CHARGE, and OUTPUT modes were tested individually to verify the operation of the charging circuit, buck converter, relays, enable button, voltage selector, indicator LEDs, OLED, and measurement system. The high temperature cut off was tested by removing the 10k Ω thermistor from its position on the battery and heating it to 95° using a heat gun. Voltage output was correctly shut off and the HIGH TEMPERATURE warning was displayed.

Output-Voltage Verification

Initial output testing began the use of a multimeter and no load at each of the 3.3 V, 5 V, and 9 V outputs. With no current for the control loop to regulate the output expectedly had significant voltage ripple but it was still clear that the system was attempting to reach each target voltage.

Figure 24: output at 9V with no load

Next the system was tested using LEDs to provide a low current load to the system. The Voltages stabilized significantly. A RS-380 DC motor was chosen after the LEDs because of its inductive load, larger 12V rating and increased current draw during start-up. The motor operated successfully at all three settings.

Figure 25: multimeter reading at 5V with load

The three output-voltage settings were measured with a digital multimeter while the DC motor was in use. The multimeter was treated as the reference measurement, and the results were compared with the nominal voltage selected by the rotary switch.

Each selectable output was measured with a digital multimeter and compared with the value reported on the OLED via the voltage divider.

Selected voltage	Measured multimeter voltage	Observed deviation from multimeter
3.3V	3.30V $\pm$ 0.04	$\pm$ 0.01 V
5V	5.00V $\pm$ 0.03	$\pm$ 0.01 V
9V	8.95V $\pm$ 0.07	$\pm$ 0.02 V

While the highest duty cycle 9V output proved to be the least efficient at an average voltage of 8.95 with an error of .78% all three outputs were well within expected tolerances under load. The observed voltage deviation in comparison to the multimeter was very small verifying a correct voltage-divider ratio, filtering, stable ADC reference, and correct firmware calibration.

Cell voltage was tested in charging mode by connecting the multimeter to differing pins of the PCB balance charger JST connector. Each of the voltages matched the displayed voltage with the least efficient reading of 2.5% error from CELL 3 due to the nature of the cell calculations from cumulative cell-tap voltages.

Current Verification

To test the current of the output voltage the multimeter was connected in series with the 380 DC motor. The resulting current deviation of the $0.05\ \Omega$ shunt and MCP6002 circuit from the displayed current was $\pm 0.02\text{A}$ which was within expected range.

Figure 26: current reading at 3.3V with load

Initially the total battery current sensor displayed a current output of 2A with output disabled which far exceeded expected current draw. This issue was partially due to an incorrect baseline calibration and mostly due to component choice. The multimeter was put in series between the positive battery terminal and the battery connector to verify current within an acceptable range further explained in the Troubleshooting and Design Corrections subsection.

The charging current sensor was tested by placing the multimeter in series between the highest voltage pin of the JST balance charge connector and the battery charger while leaving all other pins between the pcb and board connected in their intended configuration to match the placement of the current sensor. The result was an error of $\pm 0.01\text{A}$ well within expected deviation.

Testing Limitations

Testing was limited by the available equipment. A programmable electronic load and oscilloscope were unavailable; therefore, output ripple, switching waveforms, conversion efficiency, transient response, maximum continuous-current capability, and full-load thermal performance were not directly characterized.

The completed tests validate the system's operating modes, selectable output voltages, voltage-measurement accuracy, display behavior, and ability to operate both low-current LED loads and a 380 DC motor. Full-load, efficiency, ripple, and endurance testing remain areas for future evaluation.

Troubleshooting and Corrections.

Buck-converter startup failure

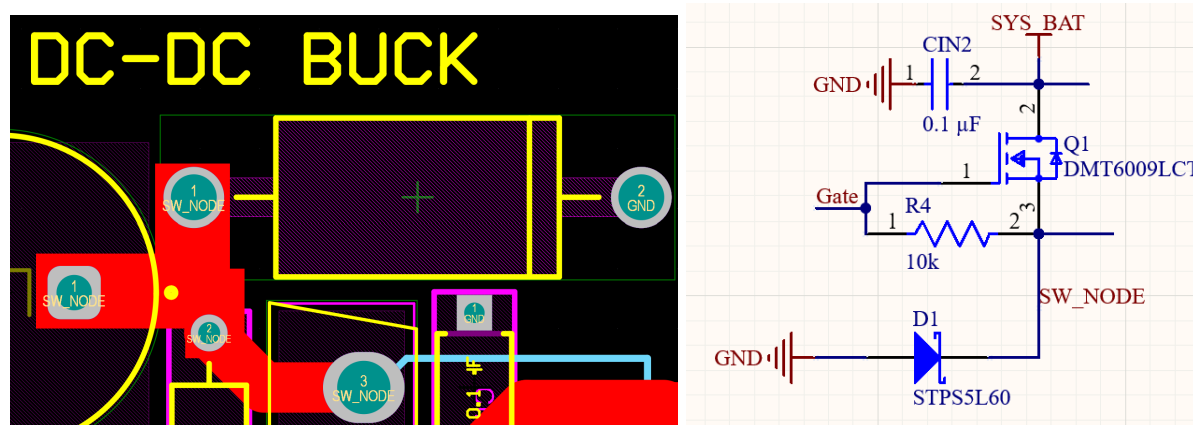


Figure 27: PCB footprint reverse of intended design

During the initial hardware bring-up, enabling the buck converter caused the 5 A battery-input fuse to immediately blow. Testing showed that the power MOSFET had failed with a short-circuit between its drain and source terminals. Replacing the MOSFET allowed further troubleshooting, but the replacement also failed when the converter was enabled. No other surrounding components appeared to be damaged indicating that the problem was within the switching-power stage rather than the firmware or control logic.

The schematic and PCB were then more thoroughly inspected and compared to verify connections. This inspection revealed that the PCB footprint for the catch diode (D1) had its anode and cathode reversed relative to the schematic symbol (Figure 27). Although the schematic showed the correct circuit, the footprint caused the physical diode to be connected with the wrong polarity while soldering components.

In the intended configuration, the catch diode's anode connects to ground and its cathode connects to the switching node. This orientation allows the diode to conduct the inductor current while the MOSFET is off. With the diode reversed it became forward-biased creating a direct short through the MOSFET to ground when the MOSFET was turned on twice damaging the component and blowing the input fuse.

After a new diode was placed in the correct orientation and a third MOSFET was installed the buck converter operated as intended and produced the selectable 3.3 V, 5 V, and 9 V outputs. This failure demonstrated that a correct schematic does not guarantee a correct physical implementation. Symbol-to-footprint pin mapping and component polarity should be verified directly against the component datasheet before manufacturing.

Total Battery-Current Measurement Limitation

The ACS758LCB-050U Hall-effect sensor used to measure total battery current was found to be poorly matched to the system's normal operating range. The sensor was originally selected for its

electrical isolation, through-hole package, and ability to tolerate high current. However, its 50 A measurement range is much larger than the approximately 0–3 A range expected during normal operation. Readings from the multimeter were used to adjust the firmware calibration of the total battery current sensor. While this helped to slightly reduce the error, official specifications show significant noise relative to sub amp measurements. The resulting current error therefore remained outside of intended range. When powered from 3.3 V, the sensor produces approximately 39.6 mV of output change per ampere.

Figure 28: Battery-current reading produced by the oversized ACS758 sensor while buck output disabled

A small offset in the zero-current reading can therefore create a large measurement error. For example, a zero-current offset of 47.5 mV results in:

$$I_{\text{error}} = 47.5 \text{ mV} / 39.6 \text{ mV per A}$$

$$I_{\text{error}} \approx 1.2 \text{ A}$$

The separate buck-output current sensor, which uses a shunt resistor and MCP6002 amplifier, produced a reading of approximately 0.4 A while operating the 380 DC motor. This reading was confirmed using a multimeter, indicating that the limitation was specific to the ACS758 battery-current measurement. A lower-range through-hole Hall-effect sensor would improve sensitivity while retaining isolation. Alternatively, a shunt-based digital current monitor such as an INA226 module would provide more accurate measurement of both the low logic current and the higher battery current present during buck-converter operation.

Conclusion

The completed project resulted in a functional prototype of a rechargeable, MSP430-controlled portable power supply. The completed electrical system is capable of providing selectable 3.3V, 5V, and 9V output from a 3S lithium-ion battery. Multimeter measurements confirmed the accuracy of the central values with a maximum percent error of .78% at an average output of 8.95V with 9V selected. Each output was successfully demonstrated using LEDs and a 380 brushed DC motor.

The OFF, OUTPUT, and CHARGE operating modes function as intended. The mode-control system prevents the battery charger and buck converter from operating simultaneously, while the enable button provides separate control of the buck converter during OUTPUT mode. The internal balance charger allows the battery to remain installed and be charged through the external charging connection.

The MSP430G2553 successfully reads and responds accordingly to the hardware selections, and manages sensor measurements, protection features, and user interface. The OLED displays battery condition, individual cell voltages, selected and measured output voltage, current measurements, temperature, and system status.

Firmware protection disables the buck converter when the output current exceeds the 2.7 A limit or when an excessive temperature is detected. The 5 A battery-input fuse and 3 A output fuse provide additional hardware protection.

All of the project's main objectives were achieved. The custom buck converter, charging system, sensing circuits, embedded firmware, four-layer PCB, and user interface were successfully integrated into one system. A custom enclosure was also developed in Onshape to organize the PCB, battery, charger, controls, display, and external wiring into a portable assembly.

The primary performance limitation is the ACS758 battery-current sensor. Its oversized 50 A range does not provide the necessary sensitivity to accurately measure the system's normal sub-ampere current. The intended 2.5 A continuous-output capability also could not be fully verified without an electronic load and oscilloscope.

Despite these limitations, the project demonstrates successful integration of power electronics, analog sensing, embedded control, PCB design, and mechanical design into a practical portable power system. The project also provided a fun and valuable opportunity to broaden my knowledge of power electronics, analog sensing, embedded programming, battery management, PCB design, hardware debugging, and mechanical CAD. Combining these disciplines to create a working prototype strengthened my system level engineering skills and increased my confidence in designing, testing and improving complex engineering projects.

References

Appendix A — Complete Electrical Schematics

This appendix contains the complete electrical schematics developed in Altium Designer for the portable power-supply system. The schematic sheets document the buck converter, embedded control system, sensing circuits, charging interface, user-interface connections, and top-level system interconnections. Cropped schematic sections are included throughout the main report, while the following pages provide the complete circuit design.

Appendix B — Complete PCB Design

Appendix C — Bill of Materials